Lightweight GNN Autoencoder for Multivariate Wearable Time-Series Anomaly Detection

Abstract

The increasing number of Internet of Things (IoT) devices, especially wearables, generates complex, multi- dimensional sensor data streams. This requires efficient and quick anomaly detection methods for applications like healthcare moni- toring and predictive maintenance. Current methods either fail to consider the relationships between different sensors, or they use computationally intensive models that aren't suitable for devices with limited resources. This study proposes an unsupervised anomaly detection framework using a lightweight Graph Neural Network (GNN) autoencoder. Sensor data is represented as a graph derived from correlations, thereby capturing structural dependencies and mitigating the need for intricate temporal architectures. This study differs from previous research by focusing on how well it applies across different subjects, using strict evaluation methods that prevent data leaks. This approach highlights the importance of evaluating models in a way that reflects real-world situations. The model's training uses only nor- mal data. It identifies anomalies by looking at the reconstruction error. Furthermore, post-training INT8 quantization improves efficiency by reducing model size and inference latency, with only a small impact on performance. This results in a compact architecture with about 3.8K parameters. We conduct thorough ablation studies to compare the proposed method with Isolation Forest, LSTM autoencoders, attention-based LSTM models, and a GNN-Transformer hybrid. Evaluations of the PAMAP2 dataset, using both between-subject and leave-one-subject-out (LOSO) methods, show strong generalization capabilities, as indicated by an ROC-AUC above 0.87. Furthermore, a scoring methodology grounded in extreme value theory (EVT) facilitates label-free assessment. The results also indicate that flawed evaluation methods, including data leakage, can lead to inflated performance metrics. In summary, these findings underscore the efficacy of lightweight graph-based modeling, thereby suggesting its appropriateness for deployment in resource-limited settings.

Index Terms:- *anomaly detection, graph neural networks, wearable sensors, edge computing, IoT, unsupervised learning, model quantization, LSTM autoencoder*